# Digital Twin ConOps as a Platform for Airport Master Planning

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***Abstract*—Airport master planning is a prediction of growth in an environmentally, operationally and physically constrained domain, which integrates reliable forecasting data with strategic business objectives in the form of phasing and land use criteria. It provides a holistic view for the short, middle and long-term development plans by mapping limitations and optioneering. This article presents digital twin (DT) concept of operations (ConOps) as a platform for drafting sustainable airport master planning framework. The idea is to have an identical twin that highlights infrastructure scalability issues and assets utilization capacity over a virtual engine, which ranges from augmented to extended reality features (AR/XR). The study assumes existence of geographical information system (GIS) geodatabase that collates data from the DT module. The GIS compares data from different sources viz. topographical surveys, as-built drawings and building information model (BIM) for ensuring consistency. The AR/XR network uses cameras, field devices and sensors for producing coordinates by virtue of industrial internet of things (IIoT) mesh. The federated model generated by DT architecture represents consolidated GIS stream with an overlay of baseline and projected requirements. The outcome is a superimposed layout that favorably validates the study hypothesis by visually displaying limitations, which may become problematic in the near future due to the provisioning of latest aviation tools and/or addition of new facilities etc., across an airport.**

***Index Terms*—digital twin, concept of operations, airport master planning, geographic information system, extended reality**

## I. Introduction

### A. Problem Statement

Modern airports are agile and resilient in terms of baggage handling, passenger processing, digital wayfinding, electronic boarding, automated border controls, dining and shopping and seamless flight operations. Considering the complex nature of an airport ecosystem, it is difficult to predict the behavior of a new facility and/or a smart system over existing infrastructure e.g. to construct a new satellite terminal, or to roll out modular data center or to commission remote air traffic control tower etc. From the master planning perspective, it is imperative to consider critical limitations such as the duct banks availability, power consumption and cooling constraints, impact of adding new fiber optic connections on network throughput, and assets utilization till date etc., with a view to provide best alternative options with minimal operational disruptions.

Air transportation has evolved and airports nowadays are focusing on a smart airport city concept for delivering unique travel experience to the passengers [1]. Luxembourg airport (LUX) has already embarked on this journey by inaugurating LUX Skypark Business Centre [2], which served as a stepping stone towards "Smart Airport 5.0". Digital transformation [3], technology upgrades and emerging trends in civil aviation are paving the way towards continuous technological revolution.

In the light of above briefing, it is critical to envisage the attributes of a system from both the operational and functional perspectives, within a virtual live environment. This is where the digital twin (DT) concept of operations (ConOps) comes in handy as it replicates the existing infrastructure and delivery of interactive situational awareness capabilities for preparing a sustainable airport master plan framework.

### B. How Digital Twin Fits in an Airport Ecosystem?

The problem statement has established the bottlenecks that need attention while master planning an airport. DT module enables airport-wide representation of utilities infrastructure, including but not limited to the system attributes, functions and behavior, utilizing augmented (AR) and extended reality (XR) features [4]. Mathematically speaking, DT is a subset of a facility's infrastructure and assets, wherein the facility is a superset of that particular DT architecture.

Hamad International Airport (HIA) in Doha, Qatar, leads in DT program and is at the forefront of adopting new trends in aviation industry. HIA's DT technology optimizes resource utilization by providing the right data at the right time, thereby reducing asset downtime and maximizing efficiency [5].

A case study of Vancouver International Airport (YVR) in British Columbia, Canada [6], has summarized the benefits of DT application across a commercial airport. YVR unveiled a development roadmap since DT has proven to be a key enabler for their digital transformation strategy.

Schiphol Airport in Amsterdam, Netherlands, is using DT for improving operational efficiency and addressing capacity challenges [7]. Hong Kong International Airport, Hong Kong, has plans for a full-scale DT unit to produce 3D replica of the physical structure and facilities, as reported in a dossier [8].



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### C. Literature Review and Aviation Industry Collaborations

IEEE published an interesting article that particularized the concept of asset twin for managing construction projects lifecycle [9]. The study documented DT system requirements but did not cover deployment scenarios. The approach was bleak since it did not investigate implications of the project or construction management, contrary to its title. The work only emphasized a notion of digital asset twin for possibly reducing the total cost of ownership (TCO) as well as enhancing human interaction for improving an asset's safety parameters.

An IEEE conference paper was presented in 2023 [10] that stated the design development of airfield lighting management system based on DT technology. The study revealed software interfaces, light intensity tests and a centralized monitoring of the basic parameters. From the conclusion, it is clear that DT paradigm has been exploited partially contrary to what has been postulated in terms of airport efficiency and reliability.

The concept of DT dashboard and dedicated databases was presented at a conference in Singapore [11]. The proposal used supervisory control and data acquisition (SCADA) for short-range, whereas long-range (LoRa) signal for outreach to field devices. This is where the discrepancy arose since LoRa needs a communication protocol for data transmission such as low-power wide area network (LoRaWAN), which was not accounted for. Despite lacking network connection details, the authors managed to develop an intriguing DT dashboard over FlexSim, a proprietary simulation modeling software.

Another extensive piece of work, published by IEEE [12], elaborated a breakdown of DT architecture into sub levels i.e. digital model (DM) and digital shadow (DS) respectively. The research focused on integrating standalone DT models into the enterprise network frame, albeit hypothetically. The question is that the master DT model will be a single point of failure, and will render the network of interconnected independent DT structures inaccessible, in case of any malfunction or security breach. Nonetheless, it was a notable effort overall.

An industrial article that was eventually published as part of a book series, discussed at length the issue of airport DT for resilient disaster management response [13]. It targeted USA airports primarily and introduced the concept of foundational twin, which acted as a master unit for interacting between the slave DT models. The main DT unit was also responsible for aggregating data from different feeds, which is analogous to the one presented above [12]. Despite the novelty of research, the risk of system failure due to malfunction or security breach persists since foundational twin is a single point of failure.

Beumer Group is an intralogistics solution provider who specializes in baggage handling system (BHS) for airports. They advanced a DT prototype for baggage management [14]. They teamed up with the aviation stakeholders and identified key performance indicators for monitoring purpose viz. power consumption, BHS mechanical and electrical alarms, system throughput, temperature and vibrations etc. Their architecture is applicable to both Greenfield and Brownfield projects.

Amadeus is an international company that offers software solution to the aviation industry. They conceptualized DT and internet of things (IoT) for transforming ground operations at an airport [15]. Their decision paper outlined benefits that will greatly improve the apron operations and reduce the fuel burn.

Airports Council International (ACI), an organization of global airport authorities for unifying standards, presented use case scenarios and applications of DT in airports [16]. Being a flagship representative of airport authorities, ACI solicited the constraints and recommended implementation zones that will optimize the operations thereby reducing delays that may occur due to planned system downtimes. ACI's roadmap will extrapolate the DT model to the urban areas periphery, which will lead to an Airport Smart City concept.

Airport Cooperative Research Program (ACRP) drafted comprehensive guidelines for DT [17], encapsulating system attributes, tiered modernization approach, business use cases, virtual reality interactions and technology barriers. ACRP is sponsored by the Federal Aviation Administration (FAA), and managed by the Transportation Research Board (TRB), USA. DT system will equip the authorities with a cohesive view of an airport's assets and infrastructure, which will facilitate the master planning process in optioneering and proposing best alternative routes for stakeholder's final approval.

## II. Digital Twin ConOps Methodology

### A. Functional and Operational Design Considerations

DT is a custom designed solution based on the conditions of an individual airport. A unified approach is not viable since there are numerous stakeholders ranging from civil aviation, airport authorities, air navigation service providers, airlines, security agencies, state regulatory bodies, emergency service providers, customs and border control etc. that need to agree on a mechanism for collecting, analyzing and sharing critical data between different users across the DT platform. Last but not the least, a twin could be either a federated system or a number of modular twins that are distributed in a master-slave configuration. The study surmise following considerations for airport master planning framework using DT ConOps:

- Aviation forecasts for DT master planning dashboard.
- Functional GIS platform for feeding the DT node.
- Secured GIS and DT nodes for receiving information.
- Aerodrome's readiness for integrating the DT system.
- Growth potential, value engineering and optimization.

Previous discussions have revealed design considerations that are bare minimum for implementing different classes of DT module across an airport. Nonetheless, the requirement for synthetically replicating all physical properties in an identical twin is illustrated below in Fig. 1. It is notable to mention that Gordon Growth Model with three variables can be applied for commercial assessment to derive a business case justification. Following airports are benchmarks of excellence that serve as a reference point for estimating the rough order of magnitude, in addition to global financial trends [18]:

- Luxembourg Airport, Findel, Luxembourg [2].
- Hamad International Airport, Doha, Qatar [5].



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- Vancouver International Airport, BC, Canada [6].
- Schiphol Airport, Amsterdam, Netherlands [7].
- Hong Kong International Airport, Hong Kong [8].

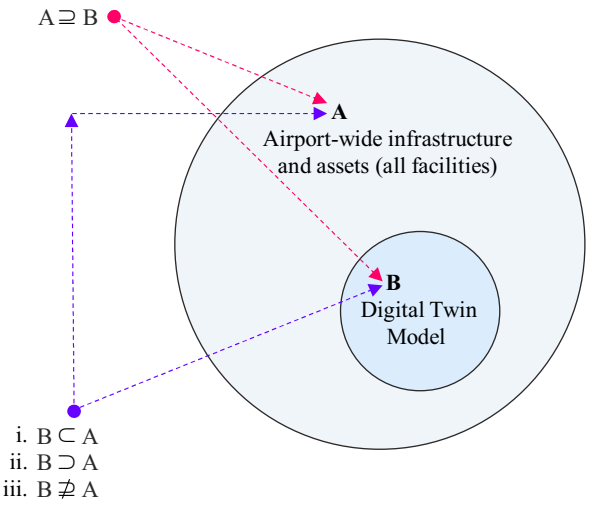



Fig. 1. Venn Diagram for an Airport and the Digital Twin Counterpart.

### B. Entity Relationship Diagram for Digital Twin ConOps

An illustration of DT ConOps covering airport-wide assets and infrastructure is shown in Fig. 2. Data from different roots feed the GIS database that correlates the information to ensure consistency of data by means of quality assurance (QA) check. GIS is the production environment whereas DT database (DB) is in the development mode. Field data synchronization occurs via sensors through an IIoT network alongside the GIS stream. DT dashboard manages the overlay from AR/XR synthesizer, highlighting all problematic areas e.g. inaccessibility of duct routes or asset capacity overutilization etc. The airport master planning framework channel these details for sustainability.

## III. High Level Overview of Digital Twin ConOps Architecture

The twin architecture for digitally displaying an overlay of airport-wide utilities and infrastructure is described in Fig. 3. For the sake of simplicity, the study considers seven reporting and relational DBs that play a vital role in orchestrating a DT network. The lower left node denotes pan, tilt and zoom (PTZ) cameras that are distributed across different facilities. Ethernet traffic transports the data over an airport-wide fiber optic (FO) network, comprising of core backbone, distribution layer and access switches. The role of AR/XR synthesizer is to generate virtual layouts by using camera coordinates and sensors data.

The lower right node aggregates sensors data through edge gateways. The criteria for edge gateways depend on the tenant and/or the airport authorities. Fog computing cloud is an ideal candidate for edge nodes [3], [19]. The choice of data protocol is directly proportional to the infrastructure readiness. For new projects, aeronautical mobile airport communications system (AeroMACS) datalink has recognition from the FAA airports engineering division [20]. AeroMACS could be a deal breaker if the cells are not present within airport's vicinity. LoRaWAN protocol will suffice the requirements if the distance between sensor nodes and the control server is less than ten kilometers. It could work up to fifteen kilometers but is not recommended. Alternatively, if the airport in question does have capacity to accommodate additional controllers, then SCADA can also be used as a cost effective strategy.

The upper right segment outlines typical processes that are necessary for concluding an airport master planning exercise. It is a recommended practice as mentioned in the latest version of airport development reference manual (ADRM), published by the International Air Transport Association (IATA) [21]. It is noteworthy to mention that Fig. 3 is for reference purposes only and does not reflect the network topology of an existing airport. In 2023, the International Civil Aviation Organization (ICAO) released an update of airport planning manual [22] that complements ADRM from master planning perspective.

The heart of core computing lies at the upper left corner that acts as a consolidation point for aggregating layouts from both the GIS and DT streams. DT dashboard extends control to user interfaces (UIs) that are not shown in the diagram since network designing is not the scope of this study. DT dashboard manages features of superimposed overlays, highlighting any bottlenecks that could be problematic in the near future, if not addressed timely. Fig. 1 enlightens the reader in understanding the reciprocity link between a DT module and an airport's facility/system for twinning the attributes, characteristics and behavior completely, under real world conditions.

## IV. Projected Results

### A. Total Cost of Ownership - Benefits Realized

The proposed architecture provides full autonomy to the business by reducing TCO in terms of directly linking the DBs i.e. point-to-point integration of relational DBs, as shown in Fig. 3. This approach is stable for commissioning DT ConOps architecture in order to formulate a sustainable, scalable and resilient airport master planning framework.

Point-to-point or peer-to-peer (P2P) integration refers to an endpoint connectivity between two relational DBs without middleware devices via a dedicated resilient link. The network topology is mesh instead of star because mesh offers higher speeds and lower latency wherein DB table entries do not need to pass through a central hub, every time a query is generated.

Enterprise service bus (ESB) has been overruled due to its integration challenges. It can act as a centralized information broker (IB) but is not recommended for linking DBs because of the maintenance requirement due to its static routing nature. They are more suitable for interfacing zero tolerance systems. The network elements of presented high level DT topology have been elaborated so far. Further validation can be done by means of a commercial or academic grade simulation engine such as Simulink (Mathworks, Natick, MA, USA), OPNET Modeler (Riverbed Technology LLC, San Francisco, CA, USA), SQL Data Base Modeler (SQLDBM), etc.



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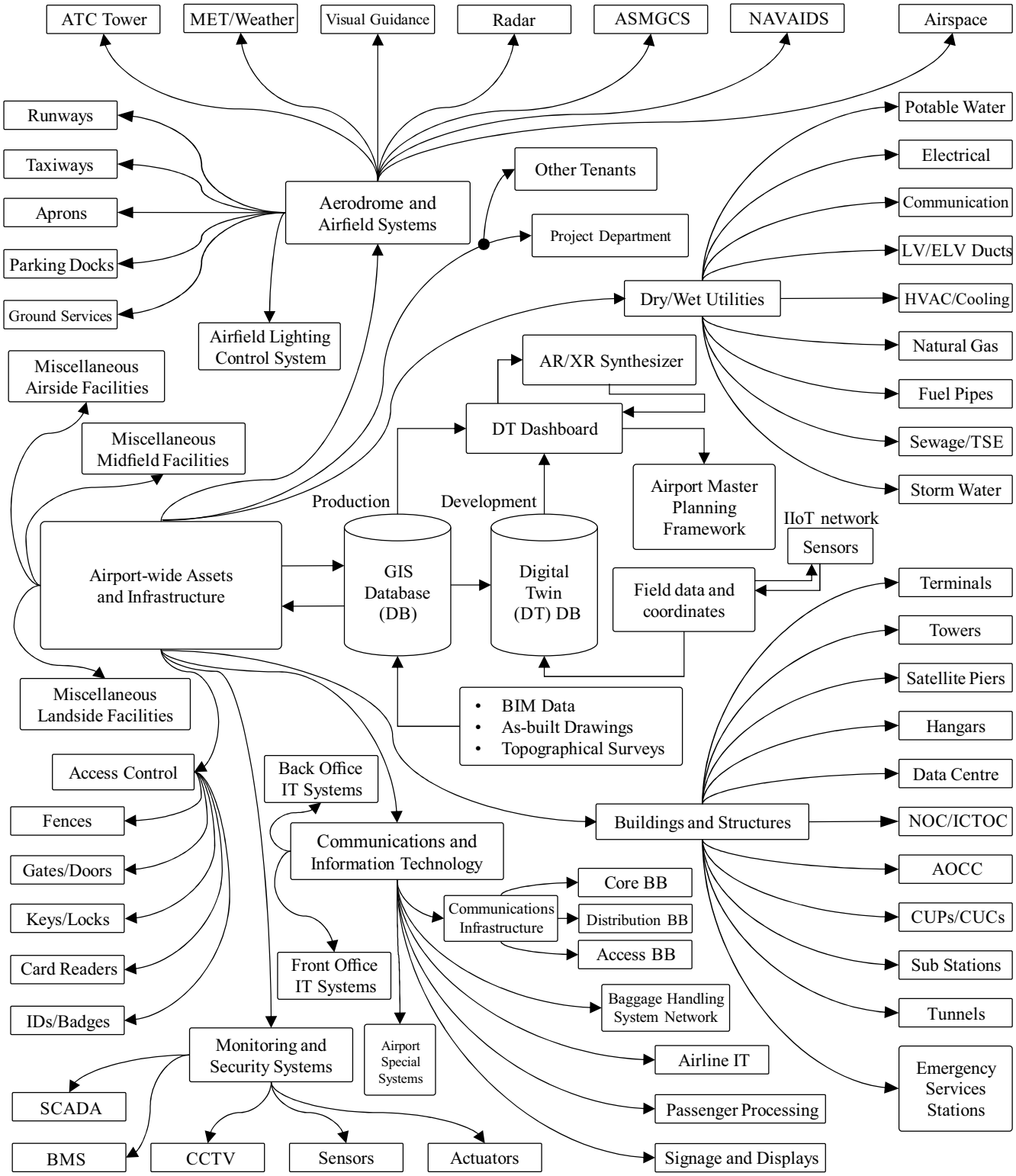



Fig. 2. Entity Relationship Diagram of Digital Twin Concept of Operations for Airport Master Planning Framework.



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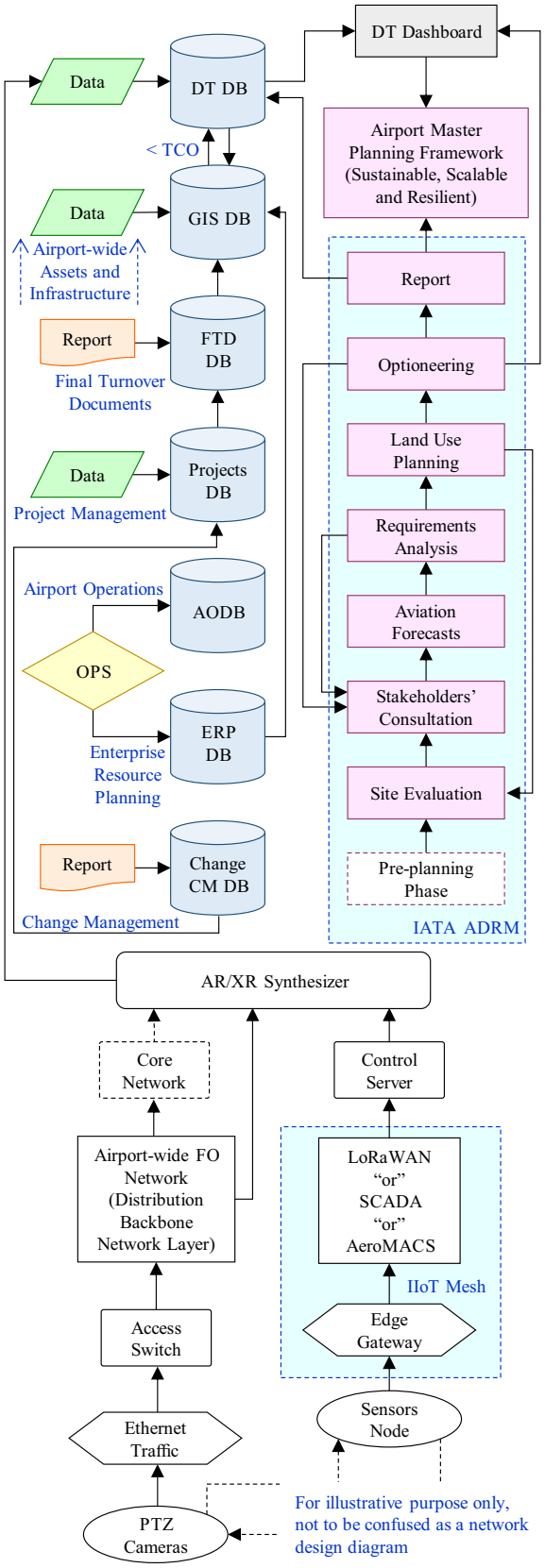



Fig. 3. High Level View of Digital Twin ConOps for Airport Master Plan.

### *B. Roadmap to Digital Transformation - Smart Airport 5.0*

Airports have undergone major technological revolutions since their inception into the air transportation model. Smart Airport 5.0 provides outlook for unmanned airport experience envisioned for travel beyond 2030, which encapsulates smart trolleys, edge computing and artificial intelligence (AI) based video analytics, unified biometric identity, facial recognition, smart toilets, electronic immigration and contactless boarding gates etc. Infrastructure readiness not only opens the door for digital transformation but also enhances the quality of airport master planning process, which justifies the study objectives.

DT ConOps architecture harness the power of emerging trends in civil aviation technology such as AR/XR, IIoT and edge computing etc. It is a customizable modular solution that can be tailored for specific applications such as scheduling for maintenance and/or roll-out activities e.g. replacing manhole lids with smart cover, investigating electrical vehicle charging faults, understanding robotic cleaning for solar panels and/or deploying the virtual control towers for air traffic management [23], etc. Quality control (QC) checks are an integral part of DT architecture that ensures consistency of data and make it a reliable model for digital transformation roadmap.

## V. Discussion

The topology proposed in this article leverages traditional master planning process, thereby reducing several iterations and daunting rounds for utilities and infrastructure re-routing and assets capacity utilization of an airport. Master planning is a thoughtful job that not only considers the requirement of internal and external stakeholders but also adopts emerging trends, aviation best practices, environmental considerations, carbon neutrality, net zero targets and offsetting of greenhouse gases emission along with the lessons learned from exercises conducted at other airports. It is a benchmarking methodology rather than a rapid ideation. Apart from resource optimization and cost reduction, a well-curated master plan conforms to the international standards and the local aviation regulations alike, whilst a data informed particularization, bespoke to the subject airport, can also be achieved deriving uniqueness rather than plagiarism. The study has corroborated this approach by virtue of DT ConOps framework as a platform for master planning.

A potential limitation of proposed topology lies in the state of current aerodrome infrastructure for different airport types all across the globe. For instance, if an airport is located on an island or on a reclaimed land, than infrastructure scaling will not be a piece of cake and master planners will have to adopt appropriate strategies from the expansion perspective. In such cases, the topology needs a bit of modification for unleashing the processing power of DT ConOps configuration. Planners can then insert air transportation forecasts, airport capacity requirements and cargo tonnage data etc. into the modified DT model. The readers need to take a note of the fact that there is no one-size-fits-all technique in today's world that is heavily driven by latest technological innovations.

Last but not the least, the presented architecture is modular in nature and DT ConOps can be extrapolated to other aviation



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domains viz., airspace planning, airport capacity optimization, airline network expansion and benchmarking programmes etc. The whole exercise has to be executed with a view of aligning business objectives to an airport's development targets, which will result in a comprehensive and sustainable master planning framework.

## VI. Conclusion

In the context of projected results, topology limitation and discussion summarized so far, it is not a far fetch conclusion to label DT ConOps as the future of airport master planning. Over the past decade, several mega projects related to airport expansion and digitalization have struggled due to either poor master planning or lack of infrastructure. Heathrow airport is a classic example of such atrocity, which is still suffering due to land acquisition issues that have been missed from its initial master plan. Al Maktoum International Airport, Dubai, United Arab Emirates, spent almost USD 85 billion till date. Yet the commencement of operations date is still in the air. Another example is Mexico City Texcoco Airport that has been cancelled and the expansion of Clark International Airport in Philippines. It is fairly common to see "partially completed," "early closed," "cancelled" or "indefinitely put on hold" sort of airport projects due to ill-considered master plan exercise.

The master plan corroborated by DT ConOps architecture delivers a distinctive and future-proof framework, which is applicable to the airside, midfield and landside developments including but not limited to air traffic control tower, passenger terminal buildings, concourses and satellite piers, outhouses, hotels and lounges, parks and business centers, hangars, cargo complexes, firefighting and airport emergency services and specialized aviation facilities. DT stack is at the forefront of civil aviation that is conceiving the prospective benefits of emerging technologies and their potential implications within an airport's ecosystem.

Possible future research include aerodrome twin, airfield ground lighting (AGL) twin, utilities and assets twin, airspace planning twin, passenger terminal twin, emergency services twin, data center twin and federated DT nodes etc. for services distribution and/or application specific deployment scenarios across an airport. This is not an exhaustive indication and DT ConOps platform should be a uniquely customized solution for each individual airport.

In conclusion, this paper calls for traditional airport master planning to be founded as a DT data platform architecture and sets out a roadmap to achieve the objective. Such a change in approach to master planning will build more confidence and certainty in both the extended operational life of assets and the transformation/expansion and addition of new infrastructure and assets as airports evolve into the new Smart Airport 5.0 operational model.

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